ΟΔΗΓΙΕΣ: Κ Σ (Σ) Λ (Λ).

ΚΕΦΑΛΑΙΟ 1: Σ - Ό - Σ (80 Ε)

1.1. Μ f A $\mathbb{R}$ A .

1.2. Δ .

1.3. Α f, g $f(x) = g(x)$ x, .

1.4. Τ $f + g$.

1.5. Τ $\dfrac{f}{g}$.

1.6. Τ $f \circ g$ f g.

1.7. Α f $1-1$, C_f .

1.8. Κ $1-1$.

1.9. Κ $1-1$.

1.10. Α $1-1$, .

1.11. Η f^{-1} C_f $x'x$.

1.12. Ο $C_f, C_{f^{-1}}$ $y = x$.

1.13. Τ f^{-1} f.

1.14. Α $\lim\limits_{x\to x_0} f(x) = \ell$, x_0.

1.15. Α $\lim\limits_{x\to x_0} f(x) = \ell$, $f(x)$ ℓ x x_0.

1.16. Η $\lim\limits_{x\to x_0} f(x)$ $f(x_0)$.

1.17. Α $\lim\limits_{x\to x_0} f(x) > 0$, $f(x) > 0$ x x_0.

1.18. Α $f(x) > 0$ x_0, $\lim\limits_{x\to x_0} f(x) > 0$.

1.19. Δ f, g, h $g(x) \leq f(x) \leq h(x)$ x x_0. Α $\lim\limits_{x\to x_0} g(x) \neq \lim\limits_{x\to x_0} h(x)$ $\lim\limits_{x\to x_0} f(x)$.

1.20. Α $\lim\limits_{x\to x_0} |f(x)| = 0$, $\lim\limits_{x\to x_0} f(x) = 0$.

1.21. Α $\lim\limits_{x\to x_0} |f(x)| = 1$, $\lim\limits_{x\to x_0} f(x) = 1$ $\lim\limits_{x\to x_0} f(x) = -1$.

1.22. Ι $\lim\limits_{x\to x_0} (f(x)g(x)) = 0$, $\lim\limits_{x\to x_0} f(x) = 0$ $\lim\limits_{x\to x_0} g(x) = 0$.

1.23. Α $\lim\limits_{x\to 0} \dfrac{x}{x} = 1$, $\lim\limits_{x\to +\infty} \dfrac{x}{x} = 1$.

1.24. Τ $P(x)$ x_0 $P(x_0)$.

1.25. Τ $+\infty$.

1.26. Α $\lim\limits_{x\to x_0} f(x) = +\infty$, $f(x) > 0$ x_0.

1.27. Α $\lim\limits_{x\to x_0} f(x) = -\infty$, $\lim\limits_{x\to x_0} |f(x)| = +\infty$.

1.28. H $x = x_0$ x_0 $+\infty$ $-\infty$.

1.29. H .

1.30. M .

1.31. H .

1.32. M .

1.33. A C_f $+\infty$, $+\infty$.

1.34. H x_0 x_0.

1.35. A x_0 x_0, x_0.

1.36. A , .

1.37. K $[a, \beta]$.

1.38. A , .

1.39. T Bolzano .

1.40. A f $[a, \beta]$ $f(a)f(\beta) < 0$, $f(x) = 0$.

1.41. A f $[a, \beta]$, $f(a)f(\beta) < 0$.

1.42. A $[a, \beta]$, $f(a)f(\beta) \geq 0$.

1.43. E f . A f Δ, Δ.

1.44. H x_0 x_0 .

1.45. T f .

1.46. A f $[a, \beta]$, $[f(a), f(\beta)]$.

1.47. T Θ E T f $f(a)$ $f(\beta)$.

1.48. A $1-1$, .

1.49. A $\lim\limits_{x \to x_0^+} f(x) \neq \lim\limits_{x \to x_0^-} f(x)$, f x_0.

1.50. M $1-1$, , .

1.51. Υ $\mathbb{R}$.

1.52. K $1-1$, .

1.53. A f $f(a)f(\beta) = 0$, $x_0 \in [a, \beta]$ $f(x_0) = 0$.

1.54. T .

1.55. A $\lim\limits_{x \to x_0} f(x) = \ell$ $\lim\limits_{x \to x_0} g(x) = +\infty$, $\lim(f(x) + g(x)) = +\infty$.

1.56. A $\lim\limits_{x \to x_0} f(x) = 0$ $\lim\limits_{x \to x_0} g(x) = +\infty$, $\lim\limits_{x \to x_0} (f(x)g(x))$ 0.

1.57. H $+\infty - \infty$.

1.58. A $\lim\limits_{x \to x_0^-} f(x) = \ell \in \mathbb{R}$, f « » x_0.

1.59. Τ $\frac{1}{x^2}$ 0 $+\infty$.

1.60. Τ $f(x) = \log(x)$ 0 $-\infty$.

1.61. Η .

1.62. Ο $f \circ g$ $g \circ f$.

1.63. Η .

1.64. Α $0 \leq f(x) \leq M$ $M \in \mathbb{R}$ x_0 $\lim\limits_{x \to x_0} [(x - x_0) f(x)] = 0$.

1.65. Α f $f(f^{-1}(x)) = x$ $x \in D_f$.

1.66. Η $f(x) = x^2$ $1-1$ $\mathbb{R}$.

1.67. Η $\ln x$ $y'y$.

1.68. Η e^x .

1.69. Τ e^x $\mathbb{R}$.

1.70. Α $f(x) < g(x)$ x_0, $\lim\limits_{x \to x_0} f(x) = \lim\limits_{x \to x_0} g(x)$.

1.71. Η x $\mathbb{R}$.

1.72. Τ $\lim\limits_{x \to x_0} f(g(x))$ $f(\lim\limits_{x \to x_0} g(x))$, $\lim\limits_{x \to x_0} g(x)$ f .

1.73. Κ .

1.74. Μ $+\infty$.

1.75. Τ $\lim\limits_{x \to 0} \frac{1}{x^{2\nu+1}}$

1.76. Η $\frac{\pm\infty}{0}$.

1.77. $f(x) = \sqrt{x}$ $x \to 0^-$ 0.

1.78. Α $\lim\limits_{x \to x_0} f(x) = l$ $\lim\limits_{x \to x_0} g(x) = m$, $l, m \in \mathbb{R}$ $f(x) < g(x)$ x_0 $l \leq m$.

1.79. Α f $f(0) < 0$ $f(x) > 0$ x, .

1.80. Κ $\mathbb{R}$ $x = 0$.

ΚΕΦΑΛΑΙΟ 2: Δ Λ (100 Ε)

2.1. Σ .

2.2. Η y x $f'(x_0)$.

2.3. Μ x_0 x_0.

2.4. Κ x_0, .

2.5. Α x_0, ΔΕΝ .

2.6. Η $|x|$ 0 0.

2.7. Η $f(x) = |x|$ 0.

2.8. H $f(x) = \sqrt{x}$ $x = 0$.

2.9. H .

2.10. O $(x_0, f(x_0))$ $f'(x_0)$.

2.11. A $x'x$, $f'(x_0) = 0$.

2.12. Δ , (. $f(x_0) = g(x_0)$ $f'(x_0) = g'(x_0)$).

2.13. K x_0 $f'(x_0) = 0$.

2.14. K .

2.15. T , .

2.16. Σ .

2.17. Θ Fermat: A f x_0 , $f'(x_0) = 0$.

2.18. A $f'(x_0) = 0$, x_0 .

2.19. A x_0 , Fermat.

2.20. A x_0 f $f'(x_0) \neq 0$, x_0 .

2.21. A f $[a, \beta]$ (a, β) $f(a) = f(\beta)$, f' (a, β).

2.22. A $\xi \in (a, \beta)$ $f'(\xi) = 0$, $f(a) = f(\beta)$.

2.23. T Θ Rolle .

2.24. T Θ.M.T. .

2.25. Σ Θ.M.T., $\xi \in (a, \beta)$ ξ AB.

2.26. T Θ.M.T. Rolle.

2.27. A $f'(x) = 0$ Δ, f Δ.

2.28. A $f'(x) = 0$, f .

2.29. A $f'(x) = g'(x)$ Δ, $f(x) = g(x) + c$ Δ.

2.30. A $f'(x) > 0$ $x \in \Delta$, f Δ.

2.31. A f , $f'(x) > 0$ x.

2.32. A , .

2.33. A ( ) . . , .

2.34. A f 1-1, .

2.35. E .

2.36. A , .

2.37. E M , M .

2.38. A $f''(x)$ x_0 ( ), x_0 .

2.39. A x_0 , $f''(x_0) = 0$.

2.40. A $f''(x_0) = 0$, x_0 .

2.41. A Δ, ” ” , .

2.42. A $f''(x) > 0$, $f(x)$.

2.43. H $(f(x) + g(x))'$ $f'(x) + g'(x)$.

2.44. O $(f(x)g(x))' = f'(x)g'(x)$.

2.45. Π 0.

2.46. H x x, ; (Ό ,).

2.47. I $(f(g(x)))' = f'(g(x)) \cdot g'(x)$.

2.48. H e^x e^x, a^x $a^x \ln a$.

2.49. H $\ln x$ $1/x$ $x > 0$, $\ln |x|$ $1/x$ $x \neq 0$.

2.50. E .

2.51. K .

2.52. O De L'Hospital $\frac{0}{0}$ $\frac{\pm\infty}{\pm\infty}$.

2.53. Γ $0 \cdot \infty$, De L'Hospital

2.54. H $\frac{|x|}{x}$ $\mathbb{R}^*$.

2.55. Υ $f'(x) = |x|$.

2.56. H .

2.57. H 3 .

2.58. H $f(x) = x^3$ $\mathbb{R}$, $x = 0$.

2.59. A f $[a, \beta]$, .

2.60. A N t, $N'(t)$.

2.61. K .

2.62. T Fermat .

2.63. A $f'(x_0) = 0$ $f''(x_0) > 0$, f x_0.

2.64. A f $x = x_0$, f x_0.

2.65. Σ , , .

2.66. H $\pm\infty$.

2.67. M ≥ 2 .

2.68. A $\lim \frac{f(x)}{x} = \lambda$ $\lim(f(x) - \lambda x) = \mu$, $y = \lambda x + \mu$ ( ) .

2.69. P 0 ( ) .

2.70. A $\lim_{x\to+\infty}(f(x) - (x + 2)) = 0$, $y = x + 2$ C_f $+\infty$.

2.71. Υ Θ M T ξ.

2.72. Γ 2, $x_1 < x_2$, .

2.73. M .

2.74. A f 1-1 , , $f'(x) \neq 0$.

2.75. H x^x $x \cdot x^{x-1}$.

2.76. A f, g , $f + g$.

2.77. T f' f.

2.78. T ( ).

2.79. A $f'(x) = 0$, f .

2.80. O λ, μ $+\infty$ $-\infty$.

2.81. A f , $f'(0) = 0$ $f(0) = 0$. (M $f(0) = 0$).

2.82. H x $(0, \pi)$.

2.83. A f x_0, $f'(x_0) = 0$.

2.84. H x_0 x_0 , $f''(x_0) = 0$.

2.85. Υ $\mathbb{R}$.

2.86. A $f''(x) > 0$ Δ, C_f Δ " " , .

2.87. $f(x) \geq f(x_0) + f'(x_0)(x - x_0)$ f.

2.88. A f, g Δ, C_f, C_g , .

2.89. T .

2.90. O .

2.91. K .

2.92. A $f''(x) > 0$, f' 1-1.

2.93. H $f'(x) > 0 \implies f$, Θ.Μ.Τ.

2.94. A $s(t)$ t, $v(t)$ $a(t) = s''(t)$.

2.95. H e^x 0 $y = x + 1$.

2.96. H $f(x) = x^3$ $x = 0$ $x'x$.

2.97. H $f(x) = \dfrac{1}{x}$ $x'x$.

2.98. K " ", .

2.99. H f $[a, b]$ a, b.

2.100. K x_0 , .

ΚΕΦΑΛΑΙΟ 3: Ο Λ (60 Ε)

3.1. Κ $[\alpha, \beta]$.

3.2. Α F, G f Δ, c, $F(x) = G(x) + c$.

3.3. Α $F(x) = G(x) + c$, F G .

3.4. Α F, G f , .

3.5. Τ $\int_a^\beta f(x)dx$, a, β .

3.6. Τ $\int_\alpha^\beta f(x)dx$.

3.7. Α $f(x) \geq 0$ $x \in [a, \beta]$, $\int_a^\beta f(x)dx \geq 0$.

3.8. Α $\int_a^\beta f(x)dx \geq 0$, f $[a, \beta]$.

3.9. Α $\int_a^\beta f(x)dx < 0$, $f(x)$ ≥ 0 $[a, \beta]$.

3.10. Α f $\int_a^\beta f(x)dx = 0$ $f(x) \geq 0$, $f(x) = 0$ $x \in [a, \beta]$.

3.11. Α - , $a < \beta$ > 0.

3.12. Ι $\int_a^\beta f(x)dx = F(\beta) - F(a)$, F f.

3.13. Τ C_f, $x'x$ $x = \alpha$, $x = \beta$, .

3.14. Υ $\mathbb{R}$.

3.15. Ι $\int_a^\beta f(t)dt + \int_\beta^a f(x)dx = 0$.

3.16. Ο x t .

3.17. Η $\left(\int_a^b f(x)dx\right)'$ $f(x)$.

3.18. Η $\int_\alpha^\beta f(x)dx$ (f), .

3.19. Τ f $x'x$ a β $E = \int_a^\beta f(x)dx$.

3.20. Α $f(x) \leq 0$ $[a, \beta]$, $-\int_a^\beta f(x)dx$.

3.21. Τ $\int_a^\beta |f(x) - g(x)|dx$.

3.22. Ο $\int_a^\beta f(x)g'(x)dx = [f(x)g(x)]_a^\beta - \int_a^\beta f'(x)g'(x)dx$. (Λ 2).

3.23. I $\int_{\alpha}^{\beta} (f(x)+g(x))dx = \int_{\alpha}^{\beta} f(x)dx + \int_{\alpha}^{\beta} g(x)dx,$ f, g $[\alpha, \beta]$.

3.24. A F f, $F'(x) = f(x)$.

3.25. A f , $\int_{-a}^{a} f(x)dx = 0.$

3.26. A f , $\int_{-a}^{a} f(x)dx = 2\int_{0}^{a} f(x)dx.$

3.27. H ” ” : $\int_{a}^{\beta} cf(x)dx = c\int_{a}^{\beta} f(x)dx.$

3.28. T .

3.29. T .

3.30. T $\int_{\alpha}^{\beta} f(x)dx$ α, β , x.

3.31. A $f(x) < g(x)$ $x \in [a, \beta]$ ( ), $\int_{a}^{\beta} f(x)dx < \int_{a}^{\beta} g(x)dx.$

3.32. ϒ $[\alpha, \beta]$ $\int_{\alpha}^{\beta} f(x)dx$.

3.33. A $\int_{a}^{\beta} f(x)dx = \int_{a}^{\beta} g(x)dx,$ $f(x) = g(x)$ $[a, \beta]$.

3.34. A $\int_{a}^{\beta} f(x)dx = 0$ f , 0. (T
> 0).

3.35. T .

3.36. T , .

3.37. T $\int_{1}^{2} e^{x}dx$ $e^2 - e$.

3.38. A f' , $\int_{a}^{\beta} f'(x)dx = f(\beta) - f(a).$

3.39. A $\int_{a}^{\beta} f(x)dx = 0$ $a \neq \beta$, f $[a, \beta]$ (f).

3.40. K .

3.41. A $\int_{\alpha}^{\beta} f(x)dx > 0$ $\alpha < \beta$, f $[\alpha, \beta]$.

3.42. H $f(x) = \ln x$ $F(x) = x\ln x - x + c$.

3.43. A f , C_f, $x'x$ $x = -\alpha, x = \alpha$ $(\alpha > 0)$
$2\int_{0}^{\alpha} |f(x)|dx.$

3.44. A $c > 0$, , $x'x$ $x = \alpha, x = \beta$ ($\alpha < \beta$)
$c(\beta - \alpha)$.

3.45. A f' , $\int_{\alpha}^{\beta} f'(x)g(x)dx = [f(x)g(x)]_{\alpha}^{\beta} - \int_{\alpha}^{\beta} f(x)g'(x)dx.$

3.46. Γ $C_f, C_g,$.

3.47. A $f(x) \geq g(x)$ $x \in [\alpha, \beta],$ $\int_{\alpha}^{\beta} (f(x) - g(x))dx.$

3.48. $\int_{a}^{b} c\,dt = c(b-a)$ $c.$

3.49. A $f'(x) = g'(x)$ $\int_{a}^{\beta} f'(x)dx = \int_{a}^{\beta} g'(x)dx \Rightarrow f(\beta) - f(a) = g(\beta) - g(a).$

3.50. I $\int_{\alpha}^{\alpha} f(x)dx = 0,$ $f.$

3.51. I $\int_{a}^{\beta} f(a)dx = f(a)(\beta - a).$

3.52. A $f(x) \geq 2,$ $\int_{1}^{3} f(x)dx \geq 4.$

3.53. A $\int_{\alpha}^{\beta} f(x)dx > 0$ $\alpha < \beta,$ $f(x) > 0$ $x \in [\alpha, \beta].$

3.54. A f $f(x) \geq 0$ $x \in [\alpha, \beta]$ $\int_{\alpha}^{\beta} f(x)dx = 0,$ $f(x) = 0$.

3.55. Γ $f,$ $\left|\int_{\alpha}^{\beta} f(x)dx\right| = \int_{\alpha}^{\beta} |f(x)|dx$ f $[\alpha, \beta].$

3.56. T $\int_{1}^{1} f(x)dx$ $f.$

3.57. A $\int_{a}^{\beta} f(x)dx = 0,$ $a = \beta.$

3.58. A $f(x) < g(x)$ $x \in [\alpha, \beta]$ (f, g), C_f, C_g $E = \int_{\alpha}^{\beta} (g(x) - f(x))dx,$ f, g .

3.59. A $\int_{\alpha}^{\beta} f(x)dx = \int_{\alpha}^{\beta} g(x)dx,$ $\alpha \neq \beta$ f, g , $[\alpha, \beta].$

3.60. T $\int_{a}^{b} |f(x)|dx$ $C_f,$ $x'x$ $x = a$ $x = b,$ $(a$ $b).$

ΑΠΑΝΤΗΣΕΙΣ (Ε)

Σ Κ Α

(Λ , Σ Λ : 1.Σ, 2.Λ .)

Ε 1: 1.Σ, 2.Λ, 3.Λ, 4.Σ, 5.Λ, 6.Λ, 7.Λ, 8.Λ, 9.Σ, 10.Σ, 11.Σ, 12.Σ, 13.Σ, 14.Λ, 15.Σ, 16.Λ, 17.Σ, 18.Λ, 19.Σ, 20.Σ, 21.Λ, 22.Λ, 23.Λ, 24.Σ, 25.Σ, 26.Σ, 27.Σ, 28.Σ, 29.Σ, 30.Λ, 31.Σ, 32.Σ, 33.Σ, 34.Λ, 35.Σ, 36.Λ, 37.Σ, 38.Λ, 39.Λ, 40.Σ, 41.Λ, 42.Σ, 43.Σ, 44.Σ, 45.Σ, 46.Σ, 47.Σ, 48.Σ, 49.Σ, 50.Σ, 51.Σ, 52.Λ, 53.Λ, 54.Σ, 55.Σ, 56.Λ, 57.Σ, 58.Λ, 59.Σ, 60.Λ, 61.Σ, 62.Σ, 63.Λ, 64.Λ, 65.Σ, 66.Σ, 67.Λ, 68.Σ, 69.Λ, 70.Σ, 71.Σ, 72.Σ, 73.Λ, 74.Σ, 75.Λ, 76.Λ, 77.Σ, 78.Σ, 79.Σ, 80.Σ

Ε 2: 1.Σ, 2.Σ, 3.Σ, 4.Λ, 5.Σ, 6.Λ, 7.Σ, 8.Λ, 9.Λ, 10.Σ, 11.Σ, 12.Σ, 13.Σ, 14.Σ, 15.Λ, 16.Λ, 17.Σ, 18.Λ, 19.Λ, 20.Σ, 21.Σ, 22.Λ, 23.Σ, 24.Σ, 25.Σ, 26.Λ, 27.Λ, 28.Σ, 29.Λ, 30.Σ, 31.Λ, 32.Σ, 33.Σ, 34.Σ, 35.Σ, 36.Σ, 37.Σ, 38.Σ, 39.Λ, 40.Λ, 41.Σ, 42.Σ, 43.Σ, 44.Λ, 45.Σ, 46.Σ, 47.Σ, 48.Σ, 49.Σ, 50.Σ, 51.Λ, 52.Σ, 53.Λ, 54.Σ, 55.Σ, 56.Σ, 57.Λ, 58.Σ, 59.Σ, 60.Σ, 61.Σ, 62.Σ, 63.Λ, 64.Σ, 65.Λ, 66.Σ, 67.Σ, 68.Σ, 69.Σ, 70.Σ, 71.Σ, 72.Σ, 73.Σ, 74.Λ, 75.Σ, 76.Λ, 77.Σ, 78.Σ, 79.Σ, 80.Σ, 81.Σ, 82.Λ, 83.Λ, 84.Λ, 85.Σ, 86.Σ, 87.Σ, 88.Σ, 89.Σ, 90.Λ, 91.Σ, 92.Σ, 93.Λ, 94.Σ, 95.Σ, 96.Σ, 97.Σ, 98.Σ, 99.Σ, 100.Σ

Ε 3: 1.Σ, 2.Σ, 3.Σ, 4.Λ, 5.Σ, 6.Λ, 7.Σ, 8.Λ, 9.Λ, 10.Σ, 11.Σ, 12.Σ, 13.Σ, 14.Σ, 15.Σ, 16.Σ, 17.Λ, 18.Λ, 19.Λ, 20.Σ, 21.Σ, 22.Λ, 23.Σ, 24.Σ, 25.Σ, 26.Σ, 27.Σ, 28.Σ, 29.Λ, 30.Λ, 31.Σ, 32.Λ, 33.Λ, 34.Λ, 35.Λ, 36.Σ, 37.Σ, 38.Σ, 39.Σ, 40.Λ, 41.Σ, 42.Σ, 43.Σ, 44.Σ, 45.Σ, 46.Σ, 47.Σ, 48.Σ, 49.Σ, 50.Σ, 51.Σ, 52.Σ, 53.Λ, 54.Σ, 55.Σ, 56.Σ, 57.Λ, 58.Σ, 59.Σ, 60.Λ